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Correlation of Warm Handoffs Versus Electronic Referrals and Engagement With Mental Health Services Co-located in a Pediatric Primary Care Clinic

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A B S T R A C T

Objective: To analyze the correlation of referral mechanism—warm handoff or electronic referral and attendance at behavioral health appointments in an outpatient pediatric primary care setting.
Methods: A retrospective cohort study was conducted in an inner-city pediatric primary care clinic from January 2019 to December 2019. Adolescent patients who screened positive for depression or anxiety were referred to a Licensed Master Social Worker (LMSW) either via a warm handoff (WH group, n = 148) or an electronic referral (EF group, n = 180). The EF group was contacted by the LMSW via telephone to schedule an appointment. Multiple logistic regression was used to analyze the correlation of type of referral, age, gender, race/ethnicity, primary language, and time between referral and first contact with attendance at three appointments.
Results: The WH group was more likely to engage with mental health services compared to the EF group (odds ratio = 3.301, 95% confidence interval = 1.850–5.902, p = .002) while age, gender, race/ethnicity, and primary language had no correlation. Within the EF group, those who were contacted by the LMSW within 3 days (1–3 days group) were more likely to attend appointments (odds ratio = 2.680, 95% confidence interval = 0.414–8.219, p = .040). There was no difference in attendance in the WH group and the 1–3 days group (p = .913)
Discussion: A warm handoff between primary care providers and behavioral health clinicians is significantly correlated with engagement with behavioral health services for adolescents who screen positive for depression or anxiety. Contact with the family within 3 days of referral is significantly correlated with engagement compared to a longer duration between referral and family contact.

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IMPLICATIONS AND CONTRIBUTION

A warm handoff between primary care providers and behavioral health clinicians is correlated with attendance at behavioral health appointments for adolescents who screen positive for depression or anxiety. Such correlation is also seen if the family is contacted within 3 days of the referral. Adherence to care is a common challenge for adolescents and it is worthwhile for a pediatric practice to take actionable steps to improve it.

Behavioral health problems are common among adolescents with depression occurring in 13.3% of the US population aged 12–17 [1], and of American adolescents aged 13–18 an estimated 31.9% had an anxiety disorder [2]. Major depressive disorder is associated with significant morbidity in adolescents. The economic burden of mental health crises leading to emergency room visits and in-patient hospitalizations [3] supports the significance of early identification and treatment of mental illness. Despite the increase in the number of child psychiatrists in recent years, between 2007 and 2010, only 27.4% of physician office visits for children and 47.9% for adolescents, resulting in a mental disorder diagnosis were to a psychiatrist [4]. This means

Conflicts of interest: The authors have no conflicts of interest to disclose.

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that the primary care pediatrician has had to adapt to the needs of their patient's mental health. However, they face many barriers including but not limited to, discomfort with their knowledge and skills, time constraints, and limited access to mental health consultation and referral resources.

A mental health care delivery model that helps mitigate some of the above barriers is the Collaborative Care model. The Collaborative Care team is led by a primary care provider (PCP) and includes behavioral health care managers, psychiatrists, and frequently other mental health professionals all empowered to work at the top of their license. Collaborative care has been shown to be better than care as usual for adolescents with depression and anxiety [5,6]. Collaborative care as a tool to promote integration of behavioral health and primary care generally calls for some degree of physical proximity between PCPs and behavioral health clinicians. Depending on the level of physical proximity such care has been labeled coordinated, colocated, and fully integrated. Coordinated model requires that behavioral health clinicians be present onsite 20%–40% of time. Co-located model has both PCPs and behavioral health clinicians located in the same physical location, but in separate departments or in unconnected physical space. Fully integrated model requires the behavioral health clinician to be a full-time provider within the primary care practice. The collaboration works both ways with PCPs having easier access to refer their patients with behavioral health concerns and supporting the behavioral health provider with medical comorbidities that may affect the behavioral health diagnosis. Over the course of the collaborative process the PCP may be able to manage mild to moderate symptoms or carry out continuation of psychiatric medication management with input from the behavioral health clinician ultimately creating access for more patients.

Establishment of collaborative care by itself is not a cure-all for improving care for patients with behavioral health problems. The model is not always successful without laying a good foundation for it [7]. This involves training staff in not only how the model functions but also why it is important to identify and manage mental illness. It also needs organizational support and funding—factors that are often beyond the control of the team attempting to set up collaborative care.

Transitions of care from one provider to another are often associated with a disruption in care which may decrease patient satisfaction and hamper the physician-patient relationship. The benefit of real-time and direct communication between referring and receiving providers has been shown to be beneficial in a variety of settings [8,9]. Such communication referred to as a "warm handoff" is particularly beneficial for services that are traditionally stigmatized or difficult to access. However, data in this field are currently limited. A systematic review of literature on warm handoffs found five studies that met their inclusion criteria [10] and finally concluded that current evidence regarding the effectiveness of warm handoffs is limited. While the use of warm handoffs has been shown to improve patient engagement [11], a study in an adult primary care setting showed that warm handoffs may not be beneficial especially in models where behavioral health care is integrated with primary care [12]. Somewhat counter-intuitively, one study on English-speaking adult Latinos found that they were four times less likely to attend an initial visit within 2 months of a referral if they received a warm handoff rather than a prescribed referral [13]. It should be noted that none of these studies were conducted in the pediatric population. There is some indication of short-term benefits of warm handoffs on patient engagement and systems utilization for pediatric primary care patients [14]. To date, there are no studies looking at the most effective mechanism of referral to behavioral healthcare services from pediatric primary care settings. This study aims to analyze the correlation of referral mechanism—warm handoff as opposed to electronic/prescribed referral- and engagement with behavioral health services in an outpatient pediatric primary care setting.

Methods

Institutional review board approval exemption was sought and obtained. This retrospective cohort study was conducted in a pediatric clinic of a large publicly funded hospital in inner-city Brooklyn, including adolescent children from ages 12 to 21 from Jan 2019 to December 2019. This clinic provides integrated mental health care [1] to over 5,000 adolescent patients annually with a Licensed Master Social Worker (LMSW) embedded in the clinic 5 days a week and a child and adolescent psychiatrist present in the clinic 1 day each week. The clinic workflow called for the patients to be screened with the Patient Health Questionnaire-9 (PHQ-9) [2,15]. General Anxiety Disorder-7 (GAD-7) [16] screen was performed for patients who reported symptoms related to an Anxiety disorder. Patients with a PHQ-9 or GAD-7 score of 10 or more were referred by the PCP to the LMSW. The referral was made either as a warm handoff (warm handoff group), which was defined as referral practice conducted between the PCP and LMSW in person with the patient (and family, if present), and the referral type (warm handoff or a referral ordered in the electronic medical record system) was used as a predictor variable. At the time of warm handoff, the LMSW performed a brief triage and connected patients to psychiatric resources, referred them to crisis intervention or scheduled an appointment for psychotherapy. Alternatively, (the control condition or e-referral group) was a referral practice where the PCP placed an electronic referral to the LMSW in the patient's electronic medical record. Within the group of patients who received an electronic referral, the period between the day of referral to when the patient was contacted by the LMSW was recorded. The period between the first contact with the LMSW and an initial appointment was noted for both groups. Demographic information including age, gender, race/ethnicity, insurance provider, and preferred language was collected to note any differences between the population being served at the clinic and patients in the study. The outcome variable of engagement was defined as attendance at greater than or equal to three visits for therapy was measured. Our LMSW tracked improvement via either PHQ-9 or GAD-7 scoring systems, with improvement defined as a PHQ-9 or GAD-7 score reduced by at least 50% or reaching a score below 10. The data log maintained by the LMSW (who was blinded to the aim of this study) was provided to the researchers in Microsoft Excel format in a de-identified manner. A retrospective analysis of data was performed to assess the correlation of referral type, age, gender, race/ethnicity, primary language, time to initial contact, and time to initial appointment and attendance at greater than or equal to three visits for therapy.

Statistical Analysis

We compared the characteristics, namely age, gender, race/ethnicity, primary language, PHQ-9 score, and GAD-7 score to

Table 1
Characteristics of patients in both referral groups

	Total n = 305	Warm handoff (n = 132)	E-referral (n = 173)	*p* value
Gender, n (%)				
Male	121 (39.6)	54(40.9)	67(38.7)	.5340
Female	184 (60.3)	78(59.1)	106(61.3)	
Age, median (range)	16 (12–21)	16 (12–21)	16 (12–21)	.0733
Language, n (%)				
English	262(85.9)	111(84.1)	151(87.3)	.235
Spanish	26(8.5)	14(10.6)	12(6.9)	
Haitian Creole	14(3.9)	5(3.8)	9(5.2)	
Bengali	3(0.1)	2(1.5)	1(0.6)	
Race/ethnicity, n (%)				
Black	273(89.5)	114(86.4)	159(91.9)	.122
Hispanic	26(8.5)	14(10.6)	12(6.9)	
Asian	6(2.0)	4(3.0)	2(1.2)	
PHQ-9 score (M ± SD)	15.72 ± 4.78	15.68 ± 4.75	15.7 ± 4.7	.970
GAD-7 (M ± SD score)	14.18 ± 2.25	14.16 ± 2.46	14.20 ± 2.48	.888

note differences between the warm handoff and e-referral groups. The primary outcome variable was engagement with services and the primary independent variable was whether the patient received a warm handoff. Other independent variables considered were age, gender, race/ethnicity, primary language, the time between referral and first contact (1–3 days, 3–5 days, 5 + days), and the time between referral and initial appointment. Variables that were statistically significant ($p < .05$) in the initial bivariable analysis or improved overall model fit were included in the final logistic regression model. Multiple logistic regression was performed to ascertain the correlation of the primary independent variable and other variables on engagement with services. The correlation of time between referral and first contact was analyzed only in the e-referral group as the first contact with LMSW was on the day of referral for the warm handoff group. Model fit was determined using Akaike information criterion and −2 log-likelihood. Data were analyzed using SAS version 9.4.

Results

During the period of data collection, 3,213 separate visits were documented with either a PHQ-9 or GAD-7 score administered. 328 referrals were made to the LMSW of which 148 patients were referred via a "warm handoff" (warm handoff group) while 180 were given a referral via our electronic medical record system (e-referral group). Of the 148 patients who received a warm handoff, 16 (10.8%) were connected to psychiatric resources or referred for crisis intervention and could not be included in the comparative analysis. Of the 180 patients in the e-referral group, 7 patients (3.9%) could not be contacted to make an appointment and also could not be included in the comparative analyses. Figure 1 is a flowchart depicting the study population selection as well as assignment to referral type.

Table 1 and Figure 2 show the characteristics of patients in each group revealing no statistical differences; 43.3% of patients received a warm handoff. The median age was 16 years; 89.5% of

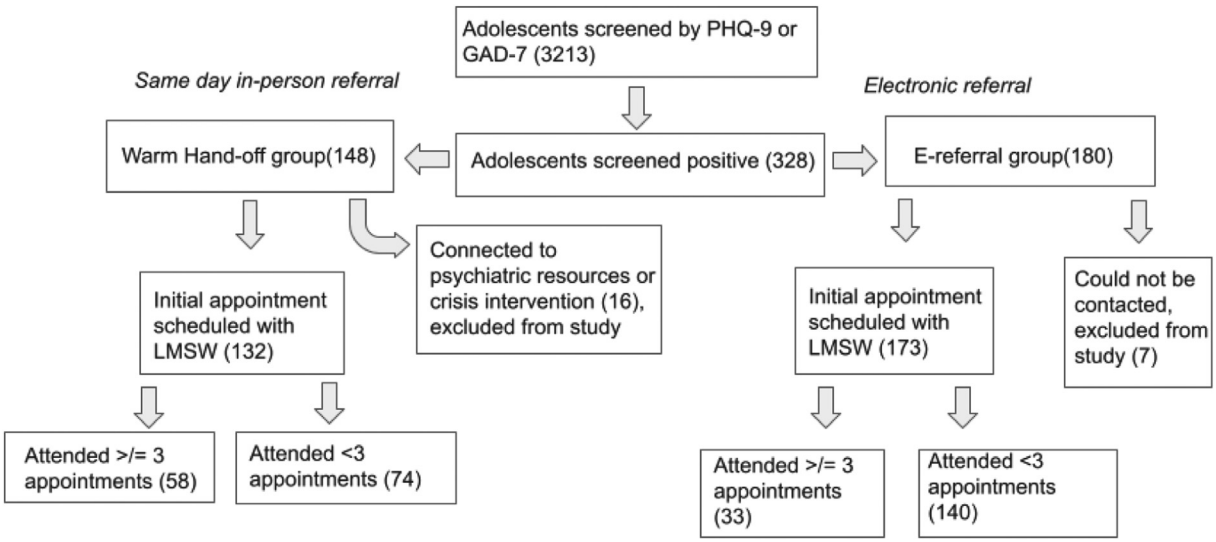

Figure 1. Study population selection and assignment to referral type.

Table 2
Correlation of predictor variables and engagement with services

	Estimate	S.E.	Wald	OR[a]	95% CI[b]		*p* value
					Lower	Upper	
Warm Hand-Off (o)	1.299	0.261	21.167	3.301	1.850	5.902	.002
Age (years)	0.057	0.055	1.056	1.06	0.952	1.182	.304
Gender (female)	0.553	0.286	3.730	1.739	0.992	3.751	.053
Race/ethnicity (Black)							
Hispanic	−0.863	0.529	2.663	0.422	0.150	1.189	.103
Asian	14.80	60.89	0.050	9.990	0.015	19.989	.983
Language (English)							
Spanish	0.899	0.665	1.827	0.177	0.667	9.059	2.458
Haitian Creole	0.990	0.932	1.129	0.129	0.433	16.437	2.692
Bengali	−14.582	68.052	0.005	0.983	0.001	99.999	.012

CI = confidence interval; OR = odds ratio; SE = standard error.
[a] Odds ratio.
[b] Confidence interval.

our study group identified as Black, 94.3% were insured by Medicaid and 60.3% were female. English was the primary language for 89.5% of the patients in the study group. The mean PHQ-9 score was 15.72 and the mean GAD-7 score was 14.18.

All patients who received a warm handoff had an initial appointment made on the same day. All initial appointments were within 4 weeks from referral for both the warm handoff and e-referral groups.

Statistical analysis showed that patients who received a warm handoff were 3 times more likely to have engagement with services while age, gender, race/ethnicity, and primary language had no statistically significant correlation (Table 2)

Within the e-referral group, those who had first contact with the LMSW within 3 days were nearly 3 times more likely to engage with services while contact in 3–5 days and more than 5 days had no significant correlation with engagement (Table 3)

There was no statistically significant difference in engagement with services in the patients in the warm handoff group and those within the e-referral group who were first contacted by the LMSW within 3 days (Table 4)

Discussion

Adherence to care in adolescents is a common challenge for healthcare providers [17]. This is reflected in our study as only 30% of the patients included in this study attended the recommended appointments. While the attendance in the e-referral group was 19.1%, even within the warm handoff group, only 44% of adolescents were attended the appointments. Rates, barriers to and predictors of adherence to care in adolescent patients have been widely studied for many chronic illnesses [18–20]. Adherence to prescribed mental health treatments and appointment attendance after psychiatric out-patient referrals has been studied in adult and mixed populations, that is both adolescents and other age groups [21–23] and some have found non-adherence to be nearly 50% [24,25]. A meta-analysis of the integrated primary care-behavioral health care model determined it to be a successful strategy for improving access to care patient outcomes and PCPs explaining behavioral services before making a referral is recommended by integrated care programs [26]. Such a model opens the possibility of warm handoffs between the primary care provider and the behavioral health provider. While warm handoffs have been shown to be beneficial in a variety of clinical settings, studies on its effect on attendance to behavioral health appointments are limited, especially for the pediatric population. Warm handoffs can be challenging due to lack of designated space and time for the meeting between providers and patients and reimbursement for services. However, the motivation behind the warm handoff is more than establishing face-to-face contact between the patient and the behavioral health provider. It also assists in conferring the rapport and trust built between the primary care provider and the patient and their family to the accepting provider which we hypothesized would positively affect engagement with recommended services. Additionally, a warm handoff may provide the behavioral health clinician to provide brief supportive counseling which may contribute to trust-building. It could also be an opportunity for the behavioral health clinician to educate the family about mental health which may help mitigate stigma attached to the diagnosis and treatment. This study attempted to analyze the correlation of the type of handoff and other factors and engagement with behavioral health services in adolescents screening positive for depression or anxiety. We found that warm handoffs, which we defined as a referral conducted

Table 3
Correlation of time between referral and first contact with family and engagement with services

	Estimate	SE	Wald	OR[a]	95% CI[b]		*p* value
					Lower	Upper	
Time to contact (>5 days)		0.419	0.092		1.110	5.570	
1–3 days	1.060	0.534	3.947	2.680	0.414	8.219	.040
3–5 days	0.394	0.648	0.369	1.482	0.416	5.298	.544

CI = confidence interval; OR = odds ratio; SE = standard error.
[a] Odds ratio.
[b] Confidence interval.

Table 4
Correlation of warm handoff versus first contact within 3 days and engagement with services

	Estimate	SE	Wald	OR[a]	95% CI[b]		*p* value
					Lower	Upper	
Warm handoff versus 1–3 days	0.584	0.574	0.122	1.068	0.148	26.454	.913

CI = confidence interval; OR = odds ratio; SE = standard error.
[a] Odds ratio.
[b] Confidence interval.

in-person on the same day, significantly correlated with attendance to three or more appointments for therapy. For the patients who did not receive a warm handoff, contact with the patient within 3 days correlated with improved attendance. Other factors like age, gender, race/ethnicity, language, and contact beyond 3 days from referral had no correlation with attendance. Notably, there was no statistically significant difference in attendance between the warm handoff group and the group with contact within 3 days. Our aim was to study the correlation of warm handoffs and engagement with behavioral health services; however, to increase generalizability we also looked at time to first contact as a variable. A possible study design to compare warm handoff versus early contact would compare appointment attendance with warm handoffs and telephone contact with the family within 24 hours.

There are several identifiable limitations of this study. The retrospective study design makes it subject to confounding. It is unclear why the specific type of referral was chosen by the PCP. None of the patients who were given an e-referral were referred for crisis intervention which may indicate that PCPs use warm handoffs for patients who have more urgent needs. Our data did not include pre-existing chronic illness in the patients, medication use, or previous experience with psychotherapy. Qualitative factors including the relationship with PCP, relationship with behavioral health clinician, the convenience of appointment time, understanding of mental illness, and necessity of treatment could all affect appointment attendance and were not studied in this research. Parental factors like motivation toward treatment, mental health diagnosis, and personal experience of mental health treatment could also affect the appointment attendance. Our study setting (urban inner-city) might lead to reduced generalizability of the results. Prospective study designs which include the above-mentioned factors with a more diverse study population will add to current research. Our literature review did not reveal a standard definition of adherence to behavioral health services, studies have used attendance to initial appointment as a measure of adherence [22,23]. We arbitrarily defined engagement with services as attendance to three appointments and our findings would likely be affected by a change in this definition.

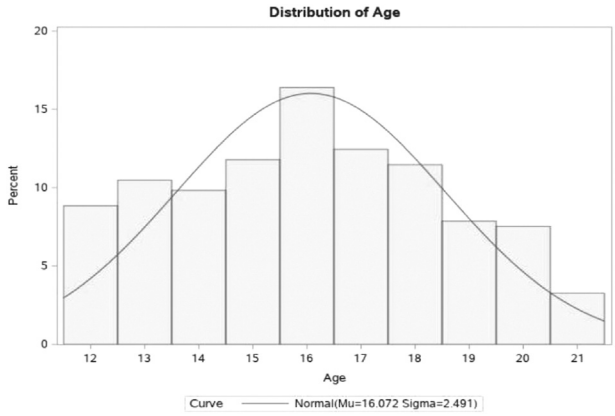

Figure 2. Age distribution of study population.

Conclusions

Our study demonstrated that warm handoff between primary care providers and behavioral health clinicians is significantly better for engagement with behavioral health services for adolescents who screen positive for depression or anxiety. Additionally, if the patient's family was contacted by the licensed master social worker within 3 days of referral, engagement was significantly better compared to a longer duration between referral and family contact. There was no statistically significant difference in engagement in patients who received a warm handoff versus those who had first contact within 3 days. While integrated behavioral health care or co-located behavioral health clinician may not be feasible for all primary care practices, workflows that limit the duration between referral and family contact should be considered.

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